

A consistent displacement Jacobian

for a proposed TPSA plasticity extension

Revised derivation and proposed changes to `Plasticity-1(1).pdf`

Status of the formulation. The elastic relations below follow the TPSA paper [1]. We retain its elastic displacement reconstruction and linear auxiliary equations, and define a plastic numerical traction using a fixed weighted transfer of cell plastic stress corrections. These are explicit *proposed discrete choices*. The Jacobian is the derivative of that specified residual on a smooth constitutive branch; spatial consistency and convergence of the plastic extension require separate verification.

1 Scope and notation

Assume small strains, a fixed mesh, fixed cellwise elastic moduli, and prescribed loads and boundary data independent of the current Newton iterate. The material example is associative J_2 plasticity. During load step $n \rightarrow n+1$, hold the committed constitutive history $\mathcal{H}_{j,n}$ and all committed cell states and face tractions at n fixed. Current candidate quantities are evaluated at $n+1$; the Newton index is omitted from the derivatives. Cell indices are a, i, j and the global face index is k .

Symbol	Meaning
$\omega_i, V_i = \omega_i $	Cell and cell measure; $i \in I_{\mathcal{T}}$.
$\zeta_k, A_k = \zeta_k $	Face and face measure; $k \in I_{\mathcal{F}}$. A face is an edge in 2D.
$\mathcal{N}_i, \mathcal{N}_k^*$	Indices of faces of cell i ; indices of cells adjacent to face k .
$\mathbf{n}_k, \mathbf{n}_{k,i}$	Fixed oriented unit face normal; outward normal relative to cell i .
$\mathbf{u}_i$	Cell displacement representing $V_i^{-1} \int_{\omega_i} \mathbf{u} dV$.
$\mathbf{r}_i, p_i$	Cell rotation-stress variable and mechanical pressure.
$\mathbf{u}, \mathbf{r}, \mathbf{p}$	Global cell vectors; independent variables before condensation.
$\boldsymbol{\sigma}, \boldsymbol{\sigma}_k$	Stress tensor; integrated numerical traction $\boldsymbol{\sigma}_k \approx \int_{\zeta_k} \boldsymbol{\sigma} \mathbf{n}_k dA \in \mathbb{R}^d$.
$\boldsymbol{\tau}_k, v_k$	Rotation-equation flux and normal displacement flux. In 3D, $\boldsymbol{\tau}_k \approx \int_{\zeta_k} \mathbf{u} \times \mathbf{n}_k dA$ and $v_k \approx \int_{\zeta_k} \mathbf{u} \cdot \mathbf{n}_k dA$.
$\overline{\mathbf{Q}}_j, \overline{\mathbf{Q}}_k$	Cell plastic stress correction and its shared numerical face value; both are second-order tensors.

The cross-product representation is three-dimensional. In reduced 2D, $\mathbf{u}_i, \boldsymbol{\sigma}_k \in \mathbb{R}^2$, and the rotation variable and rotation flux are scalar. Use the corresponding reduced blocks. A 2D material model must also specify its plane-strain or plane-stress treatment.

2 Elastic TPSA relations and strain reconstruction

For an interior face, TPSA with the paper's internal rotation-interpolation choice $\alpha = 0$ gives [1, Eq. (3.7)]

$$\begin{bmatrix} \boldsymbol{\sigma}_k^e \\ \boldsymbol{\tau}_k^e \\ v_k^e \end{bmatrix} = A_k \begin{bmatrix} -2\bar{\mu}_k \delta_k^{-1} \Delta_k^* & -R_k^n \tilde{\Xi}_k & \mathbf{n}_k \tilde{\Xi}_k \\ -R_k^n \Xi_k & 0 & 0 \\ \mathbf{n}_k^\top \Xi_k & 0 & -\delta_k^\mu \Delta_k^* \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ \mathbf{r} \\ \mathbf{p} \end{bmatrix}. \quad (1)$$

Here the superscript e identifies the elastic reference flux. For $\mathcal{N}_k^* = \{i, j\}$, order the neighbors so that $\mathbf{n}_k = \mathbf{n}_{k,i} = -\mathbf{n}_{k,j}$. Then

$$\begin{aligned} \Delta_k^* \mathbf{u} &= \mathbf{u}_i - \mathbf{u}_j, & \delta_k &= \delta_k^i + \delta_k^j, \\ \Xi_{k,i} &= \frac{\mu_i / \delta_k^i}{\mu_i / \delta_k^i + \mu_j / \delta_k^j}, & \tilde{\Xi}_{k,i} &= 1 - \Xi_{k,i}, \\ \delta_k^\mu &= \left(\frac{2\mu_i}{\delta_k^i} + \frac{2\mu_j}{\delta_k^j} \right)^{-1}, & \bar{\mu}_k &= \frac{\delta_k}{\delta_k^i / \mu_i + \delta_k^j / \mu_j}. \end{aligned} \quad (2)$$

Both averaging weights are zero outside $\mathcal{N}_k^*$. On an interior face, $\Xi_k = \tilde{\Xi}_k$ exactly when $\mu_i / \delta_k^i = \mu_j / \delta_k^j$. In 3D, $R_k^n = S^* \mathbf{n}_k$ acts as $R_k^n \mathbf{w} = \mathbf{n}_k \times \mathbf{w}$. It is a skew cross-product map and satisfies $(R_k^n)^2 = \mathbf{n}_k \mathbf{n}_k^\top - I_3$. Consequently,

$$\begin{aligned} \boldsymbol{\tau}_k^e &= -A_k R_k^n \Xi_k \mathbf{u}, & v_k^e &= A_k \left(\mathbf{n}_k^\top \Xi_k \mathbf{u} - \delta_k^\mu \Delta_k^* \mathbf{p} \right), \\ A_k \mathbf{u}_k &= R_k^n \boldsymbol{\tau}_k^e + v_k^e \mathbf{n}_k, \\ \mathbf{u}_k &= \left[-(R_k^n)^2 + \mathbf{n}_k \mathbf{n}_k^\top \right] \Xi_k \mathbf{u} - \delta_k^\mu (\Delta_k^* \mathbf{p}) \mathbf{n}_k \\ &= \Xi_k \mathbf{u} - \delta_k^\mu (\Delta_k^* \mathbf{p}) \mathbf{n}_k. \end{aligned} \quad (3)$$

Equation (3) retains the elastic TPSA reconstruction in the proposed plastic scheme. With the chosen boundary reconstruction, define the total small-strain approximation

$$\boldsymbol{\varepsilon}_j^{\text{rec}} = \frac{1}{2V_j} \sum_{k \in \mathcal{N}_j} A_k (\mathbf{u}_k \otimes \mathbf{n}_{k,j} + \mathbf{n}_{k,j} \otimes \mathbf{u}_k) = f_j(\mathbf{u}, \mathbf{r}, \mathbf{p}). \quad (4)$$

Interior contributions depend on $\mathbf{u}, \mathbf{p}$; boundary reconstruction may introduce additional dependence. On a prescribed displacement face, $\mathbf{u}_k = \mathbf{g}_k$ and its variation is zero within a fixed load step. The cell-to-face assignment is

$$\boldsymbol{\varepsilon}_{j,k} = \boldsymbol{\varepsilon}_j^{\text{rec}}, \quad j \in \mathcal{N}_k^*. \quad (5)$$

This does not identify the two neighboring strains with one another.

With exact face averages, (4) is the divergence-theorem identity for the average symmetric gradient. On affine triangles and tetrahedra it is also the constant strain of the vector Crouzeix–Raviart function with those face-average degrees of freedom [3]. Each face carries d scalar displacement components. This equivalence does not identify TPSA with a CR discretization or establish its reconstruction accuracy.

3 Force balance and cell constitutive update

Use the TPSA convention $\nabla \cdot \boldsymbol{\sigma} = \mathbf{f}$. Integration over cell ω_a gives

$$\int_{\partial\omega_a} \boldsymbol{\sigma} \mathbf{n} dA = \int_{\omega_a} \mathbf{f} dV =: \mathbf{b}_a. \quad (6)$$

Define the cell-face incidence by

$$\Delta_{a,k} = \begin{cases} +1, & \mathbf{n}_k \text{ points outward from } \omega_a, \\ -1, & \mathbf{n}_k \text{ points inward to } \omega_a, \\ 0, & k \notin \mathcal{N}_a. \end{cases} \quad (7)$$

The numerical force residual is

$$\mathbf{res}_a = \sum_{k \in \mathcal{N}_a} \Delta_{a,k} \boldsymbol{\sigma}_{k,n+1} - \mathbf{b}_a. \quad (8)$$

The integrated traction already contains A_k . It has opposite signs in the residuals of its two adjacent cells.

At fixed committed history, use the reconstructed total strain as input:

$$\boldsymbol{\sigma}_j^{\text{tri}} = \mathbb{C}_j^e : (\boldsymbol{\epsilon}_{j,n+1}^{\text{rec}} - \boldsymbol{\epsilon}_{j,n}^p), \quad (9)$$

$$\boldsymbol{\sigma}_j^{\text{return}} = \mathbb{C}_j^e : [\boldsymbol{\epsilon}_{j,n+1}^{\text{rec}} - \boldsymbol{\epsilon}_j^p(\boldsymbol{\epsilon}_{j,n+1}^{\text{rec}}; \mathcal{H}_{j,n})]. \quad (10)$$

For a consistent committed state, $\boldsymbol{\sigma}_j^{\text{tri}} = \boldsymbol{\sigma}_{j,n} + \mathbb{C}_j^e : (\boldsymbol{\epsilon}_{j,n+1}^{\text{rec}} - \boldsymbol{\epsilon}_{j,n}^{\text{rec}})$. The cell stress tensor and the numerical integrated face traction are distinct quantities in this construction.

For associative J_2 plasticity, define the 3D deviator and equivalent stress by

$$\mathbf{s} = \boldsymbol{\sigma} - \frac{1}{3} \text{tr}(\boldsymbol{\sigma}) I_3, \quad q = \sqrt{\frac{3}{2}} \|\mathbf{s}\|, \quad \Phi = q - \sigma_y(\kappa), \quad \mathbf{N} = \frac{\partial \Phi}{\partial \boldsymbol{\sigma}} = \sqrt{\frac{3}{2}} \frac{\mathbf{s}}{\|\mathbf{s}\|}. \quad (11)$$

The norm is Frobenius. On the active plastic branch, the backward-Euler flow increment is $\Delta \boldsymbol{\epsilon}_j^p = \Delta \gamma_j \mathbf{N}_{j,n+1}$; the local return map supplies the multiplier and hardening update. On an elastic branch $\Delta \gamma_j = 0$, so a flow direction at $\mathbf{s} = 0$ is not needed. The chosen 2D reduction must retain the corresponding 3D material behavior.

Define the *incremental* plastic stress correction and algorithmic tangent:

$$\overline{\mathbf{Q}}_j = \mathbb{C}_j^e : (\boldsymbol{\epsilon}_{j,n+1}^p - \boldsymbol{\epsilon}_{j,n}^p) = \Delta \gamma_j \mathbb{C}_j^e : \mathbf{N}_{j,n+1} = \boldsymbol{\sigma}_j^{\text{tri}} - \boldsymbol{\sigma}_j^{\text{return}}, \quad (12)$$

$$\mathbb{C}_j^{\text{alg}} = \left. \frac{\partial \boldsymbol{\sigma}_j^{\text{return}}}{\partial \boldsymbol{\epsilon}_j^{\text{rec}}} \right|_{\mathcal{H}_{j,n}}, \quad \frac{\partial \overline{\mathbf{Q}}_j}{\partial \boldsymbol{\epsilon}_j^{\text{rec}}} = \mathbb{C}_j^e - \mathbb{C}_j^{\text{alg}}. \quad (13)$$

The last identity includes variations of both $\Delta \gamma_j$ and $\mathbf{N}_{j,n+1}$. It differentiates the current total strain, with history and elastic modulus fixed. On an elastic branch $\mathbb{C}_j^{\text{alg}} = \mathbb{C}_j^e$. For the constant extension (5), the one-sided face average $\overline{\mathbf{Q}}_{j,k} = A_k^{-1} \int_{\zeta_k} \overline{\mathbf{Q}}_j dA$ is simply $\overline{\mathbf{Q}}_j$.

4 Shared numerical traction and auxiliary elimination

For an interior face, choose fixed transfer weights and define

$$\boxed{\bar{\mathbf{Q}}_k = \sum_{j \in \mathcal{N}_k^*} \alpha_{kj} \bar{\mathbf{Q}}_j, \quad \sum_{j \in \mathcal{N}_k^*} \alpha_{kj} = 1.} \quad (14)$$

Nonnegative weights give a convex transfer. The normalized transfer preserves a constant correction and supplies one shared face value; normalization alone does not prove spatial consistency. These α_{kj} are transfer weights, distinct from the TPSA internal rotation-interpolation parameter $\alpha = 0$ in Section 2.

Let $\mathbf{y} = [\mathbf{u}, \mathbf{r}, \mathbf{p}]^\top$, and write the first row of (1) as

$$T_k \mathbf{y} = T_{k,u} \mathbf{u} + T_{k,r} \mathbf{r} + T_{k,p} \mathbf{p}. \quad (15)$$

Define the current trial traction and the proposed updated numerical traction:

$$\boldsymbol{\sigma}_{k,n+1}^{\text{tri}}(\mathbf{y}) = \boldsymbol{\sigma}_{k,n} + T_k(\mathbf{y} - \mathbf{y}_n). \quad (16)$$

$$\boxed{\boldsymbol{\sigma}_{k,n+1}(\mathbf{y}) = \boldsymbol{\sigma}_{k,n+1}^{\text{tri}}(\mathbf{y}) - A_k \bar{\mathbf{Q}}_k(\mathbf{y}) \mathbf{n}_k.} \quad (17)$$

The correction is incremental because $\boldsymbol{\sigma}_{k,n}$ already includes previously committed plasticity. The current $T_k(\mathbf{y} - \mathbf{y}_n)$ changes during Newton iteration. An initial elastic predictor can be reused only for an identical reference operator and complete incremental right-hand side, including boundary data.

Equation (17) is a *definition of the proposed numerical flux*, motivated by the continuum stress split. Repeating TPSA's original one-sided traction elimination with a plastic stress term may also change the face displacement and auxiliary fluxes. The present proposal explicitly retains (3) and the linear auxiliary equations below.

For the reference TPSA block system, assume the selected boundary treatment gives

$$\begin{bmatrix} A_{uu} & A_{ur} & A_{up} \\ A_{ru} & A_{rr} & 0 \\ A_{pu} & 0 & A_{pp} \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ \mathbf{r} \\ \mathbf{p} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_u \\ \mathbf{f}_r \\ \mathbf{f}_p \end{bmatrix}. \quad (18)$$

Retain its rotation and pressure equations in the proposed plastic scheme:

$$A_{rr} \mathbf{r}(\mathbf{u}) = \mathbf{f}_r - A_{ru} \mathbf{u}, \quad A_{pp} \mathbf{p}(\mathbf{u}) = \mathbf{f}_p - A_{pu} \mathbf{u}. \quad (19)$$

The right-hand sides include integrated sources and boundary contributions. For fixed matrices and fixed $\mathbf{f}_r, \mathbf{f}_p$ within the load step,

$$\mathbf{R} := D_{\mathbf{u}} \mathbf{r} = -A_{rr}^{-1} A_{ru}, \quad \mathbf{P} := D_{\mathbf{u}} \mathbf{p} = -A_{pp}^{-1} A_{pu}. \quad (20)$$

Interpret the inverses as linear solves on constrained spaces; handle a pressure nullspace when present. A_{pp} is generally not diagonal because of the pressure stabilization. If the actual auxiliary equations or boundary couplings are changed, differentiate and condense that actual system.

Boundary faces. Use the selected boundary traction law and its inhomogeneous contributions. A prescribed traction independent of displacement has zero derivative. On a displacement boundary, the face displacement is fixed, while the reaction traction is unknown. Interior flux formulas do not replace the necessary boundary treatment.

5 Condensed face Jacobian and global assembly

Define the condensed reconstruction and its derivative by

$$\begin{aligned}\boldsymbol{\varepsilon}_j^{\text{rec},c}(\mathbf{u}) &= f_j(\mathbf{u}, \mathbf{r}(\mathbf{u}), \mathbf{p}(\mathbf{u})), \\ \mathcal{B}_{j,i}^c &:= \frac{\partial \boldsymbol{\varepsilon}_j^{\text{rec},c}}{\partial \mathbf{u}_i} = \left(\frac{\partial f_j}{\partial \mathbf{u}_i} \right)_{r,p} + \frac{\partial f_j}{\partial \mathbf{r}} \mathbf{R}_i + \frac{\partial f_j}{\partial \mathbf{p}} \mathbf{P}_i,\end{aligned}\quad (21)$$

where $\mathbf{R}_i$ and $\mathbf{P}_i$ denote the global matrix block columns corresponding to cell displacement $\mathbf{u}_i$. For example, an interior face varies as

$$\delta \mathbf{u}_k = \Xi_k \delta \mathbf{u} - \delta_k^\mu \mathbf{n}_k \Delta_k^* \mathbf{P} \delta \mathbf{u}. \quad (22)$$

Substitute this into (4) to include pressure feedback. The rotation term in (21) is zero if the selected reconstruction is independent of $\mathbf{r}$, including at boundary faces.

Define $\boldsymbol{\sigma}_k^c(\mathbf{u}) = \boldsymbol{\sigma}_{k,n+1}(\mathbf{u}, \mathbf{r}(\mathbf{u}), \mathbf{p}(\mathbf{u}))$ and $T_{k,u_i} := \partial(T_{k,u} \mathbf{u}) / \partial \mathbf{u}_i$. The trial-traction derivative is

$$\mathbf{J}_{k,i}^{\text{tri},c} = \frac{\partial \boldsymbol{\sigma}_{k,n+1}^{\text{tri},c}}{\partial \mathbf{u}_i} = T_{k,u_i} + T_{k,r} \mathbf{R}_i + T_{k,p} \mathbf{P}_i. \quad (23)$$

The face Jacobian, acting on a displacement variation, is

$$\boxed{\begin{aligned}\mathbf{J}_{k,i} \delta \mathbf{u}_i &= \mathbf{J}_{k,i}^{\text{tri},c} \delta \mathbf{u}_i \\ &\quad - A_k \sum_{j \in \mathcal{N}_k^*} \alpha_{kj} \left[(\mathbb{C}_j^e - \mathbb{C}_j^{\text{alg}}) : (\mathcal{B}_{j,i}^c \delta \mathbf{u}_i) \right] \mathbf{n}_k.\end{aligned}} \quad (24)$$

This is the complete derivative of (17) for fixed weights, geometry, history, elastic moduli, and the retained auxiliary equations. Do not cancel its elastic trial term against the $\mathbb{C}_j^e$ term: no identity has been established equating the TPSA elastic traction with a weighted traction of the reconstructed cell elastic stresses. If the weights are state-dependent, additionally include $-A_k \sum_j (\delta \alpha_{kj}) \bar{\mathbf{Q}}_j \mathbf{n}_k$.

For the condensed cell residual, the global blocks are

$$\boxed{\mathbf{J}_{a,i}^{\text{glob}} = \frac{\partial \text{res}_a^c}{\partial \mathbf{u}_i} = \sum_{k \in \mathcal{N}_a} \Delta_{a,k} \mathbf{J}_{k,i}, \quad a, i \in I_{\mathcal{T}}.} \quad (25)$$

The orientation index is the residual cell a . The displacement column i can range over all cells; a block vanishes only where the full dependency is absent. The adjacent index j in (24) identifies a cell supplying a return-map correction, not the range of i . For face-major and cell-major ordering,

$$\mathbf{J}^{\text{glob}} = (\Delta \otimes I_d) \mathbf{J}^{\text{face}}, \quad \mathbf{J}^{\text{face}} \in \mathbb{R}^{dN_f \times dN_c}, \quad \mathbf{J}^{\text{glob}} \in \mathbb{R}^{dN_c \times dN_c}. \quad (26)$$

There is no additional A_k in this assembly. In 3D each face block is 3×3 ; in a 2D $M \times N$ cell mesh the unconstrained displacement Jacobian is $2MN \times 2MN$.

At each Newton iterate reconstruct strains, evaluate return maps from the same $\mathcal{H}_n$, form the current residual and Jacobian, and solve $\mathbf{J}^{\text{glob}} \delta \mathbf{u} = -\text{res}^c$. Commit material history and face tractions only after convergence. Check the derivative by comparing $\mathbf{J}^{\text{glob}} \mathbf{h}$ with a centered residual difference, recomputing auxiliaries and return maps for every perturbation. Avoid a yield switch when assessing smooth-branch derivative accuracy. Verify the elastic limit, interior-face cancellation, and mesh convergence separately.

6 Proposed changes relative to the submitted draft

This table refers to equation numbers in the submitted `Plasticity-1(1).pdf`. Equation numbers in the revised derivation are new; references below link to the replacement formulas.

Submitted location	Proposed change
Notation table	Replace the face-measure denominator in the displacement average by $ \omega_i $. Distinguish stress tensors from integrated traction vectors and clarify the 3D/reduced-2D rotation convention.
Eqs. (1)–(5)	Show the normal contraction explicitly; define $\bar{\mu}_k$ and δ_k^μ ; retain the corrected orientation, skew-map identity, and pressure sign. See (1)–(3).
Eq. (7), Section 1.2	Correct the range index and identify cellwise extension as a reconstruction choice. State the affine-simplex CR interpretation with d scalar components per face.
Eqs. (15)–(17), (27)	Separate the initial elastic predictor from the current trial traction. Define the plastic face flux explicitly and use an incremental plastic correction. See (12), (16)–(17).
Eqs. (19)–(21)	Retain the cellwise-constant reduction, repair the transfer indices, state fixed normalized weights, and use one shared correction for each interior face. See (14).
Eqs. (22), (28)	Include condensation feedback inside the strain derivative and the elastic trial derivative. Use the local identity $D_\epsilon \bar{\mathbf{Q}} = \mathbb{C}^e - \mathbb{C}^{\text{alg}}$. See (21)–(24).
Eq. (23)	Replace $\Delta_{i,k}$ by $\Delta_{a,k}$ and allow all displacement columns $i \in I_{\mathcal{T}}$. See (25).
Eqs. (24)–(25)	State that the elastic auxiliary equations are retained by choice. Use their exact derivatives and handle constrained spaces and boundary contributions. See (18)–(20).
FEM comparison and boundaries	Remove the claimed contrast about solving displacement increments. Differentiate boundary flux laws separately; prescribed Neumann traction has zero derivative for fixed loading.

Accuracy and consistency wording. The divergence theorem gives an exact average-gradient identity when the face averages are exact. Syrakos et al. [2] show that the common Green–Gauss variant studied is second-order on structured meshes but can be zeroth-order on general unstructured meshes; the least-squares variants studied are second-order and first-order, respectively. This does not determine the accuracy of the pressure-corrected TPSA reconstruction. An exact Jacobian of the chosen residual also does not establish spatial consistency or unconditional quadratic Newton convergence.

References

- [1] J. M. Nordbotten and E. Keilegavlen. *Two-point stress approximation: A simple and robust finite volume method for linearized (poro-)elasticity and Stokes flow*. *Computers & Mathematics with Applications* 197 (2025), 259–294. doi:10.1016/j.camwa.2025.07.035. Detailed equation references follow arXiv:2405.10390v3.
- [2] A. Syrakos, S. Varchanis, Y. Dimakopoulos, A. Goulas, and J. Tsamopoulos. *A critical analysis of some popular methods for the discretisation of the gradient operator in finite volume methods*. *Physics of Fluids* 29 (2017), 127103. doi:10.1063/1.4997682.
- [3] M. Crouzeix and P.-A. Raviart. *Conforming and nonconforming finite element methods for solving the stationary Stokes equations*. *RAIRO Analyse numérique* 7(R3) (1973), 33–75. doi:10.1051/m2an/197307R300331.

A ADDED: restricted scope and required inputs

Reading guide. Pages 1–6 reproduce the previous revised document, `Plasticity_revised_oct6.pdf`, as the black-text baseline. All material from this page onward is new, printed in dark blue, and marked **ADDED**. The present restriction supersedes the general boundary discussion in the baseline. Equation numbers D1 onward belong only to these additions.

Full prescribed displacement on the entire external boundary. Every boundary face has all displacement components prescribed:

$$\mathbf{u}_k = \mathbf{g}_{k,n+1}, \quad \delta \mathbf{u}_k = 0, \quad k \in I_{\mathcal{F}}^D. \quad (\text{D1})$$

Here $\mathbf{g}_{k,n+1}$ is the chosen approximation to the boundary-face average. It is fixed during a Newton solve but may change between loading steps. Write $I_{\mathcal{F}}^{\text{int}}$ for the interior-face indices, so $I_{\mathcal{F}} = I_{\mathcal{F}}^{\text{int}} \cup I_{\mathcal{F}}^D$. There are no Neumann, Robin, or componentwise mixed faces in this restricted proposal. Orient every boundary normal outward: $\mathbf{n}_k = \mathbf{n}_{k,j}$ for the unique adjacent physical cell j .

The prescribed quantity is a *face displacement*, not the neighboring cell average. The cell unknown $\mathbf{u}_j$ remains an unknown. In particular, do not set $\mathbf{u}_j = \mathbf{g}_k$ or delete its Newton column solely because cell j touches a Dirichlet face.

Material and geometry inputs. Use the same fixed isotropic elastic parameters in the TPSA reference operator and the local constitutive update. For a symmetric 3D tensor $\mathbf{A}$,

$$\mathbb{C}_j^e : \mathbf{A} = 2\mu_j \mathbf{A} + \lambda_j \text{tr}(\mathbf{A}) \mathbf{I}_3. \quad (\text{D2})$$

Supply $V_j, A_k, \mathbf{n}_k, \mathbf{n}_{k,j}$ and positive normal distances $\delta_k^j = (\mathbf{x}_k - \mathbf{x}_j) \cdot \mathbf{n}_{k,j}$. The material routine must supply the chosen hardening law, returned stress, updated history candidate, and its algorithmic tangent at fixed $\mathcal{H}_{j,n}$. The tangent must differentiate that same returned-stress routine. For a 2D implementation, specify a consistent reduction. In plane strain, embed the reconstructed total strain with $\varepsilon_{33} = \varepsilon_{13} = \varepsilon_{23} = 0$, while retaining the required 3D constitutive stresses and plastic history. Use matching tensor component order and scaling, including Mandel scaling if a tensor-to-vector encoding is used.

Fixed transfer weights are supplied scheme inputs. Before computation, specify and store every interior-face weight α_{kj} in baseline equation (14), with $\sum_{j \in \mathcal{N}_k^*} \alpha_{kj} = 1$. Their numerical values, not merely their normalization, are required inputs. This supplement does not prescribe an unverified coefficient-dependent rule. For a Dirichlet boundary face, the new one-sided plastic correction below uses only its adjacent physical cell.

Reconstruction simplification. Under (D1) and the retained interior reconstruction (3),

$$\boldsymbol{\varepsilon}_j^{\text{rec}} = f_j(\mathbf{u}, \mathbf{p}; \mathbf{g}_{n+1}), \quad \left(\frac{\partial f_j}{\partial \mathbf{r}} \right)_{u,p,g} = 0. \quad (\text{D3})$$

This removes the direct rotation term from the strain chain rule. Rotation still enters the elastic traction and its condensed derivative.

B ADDED: Dirichlet reaction traction and load-data terms

Let boundary face k have unique physical neighbor j , and define $a_k = 2\mu_j/\delta_k^j$. The elastic TPSA one-sided boundary law is

$$\boldsymbol{\sigma}_k^e = A_k [a_k(\mathbf{g}_k - \mathbf{u}_j) - R_k^n \mathbf{r}_j + p_j \mathbf{n}_k]. \quad (\text{D4})$$

The signs follow the outward normal convention and $(S^* \mathbf{r}_j) \mathbf{n}_k = -R_k^n \mathbf{r}_j$; see TPSA Appendix A2.2, in particular Eq. (A2.26) [1]. The displayed cross-product blocks are 3D; a 2D implementation uses their consistent reduced form.

Define load-step differences $\Delta \mathbf{g}_k = \mathbf{g}_{k,n+1} - \mathbf{g}_{k,n}$, $\Delta \mathbf{u}_j = \mathbf{u}_j - \mathbf{u}_{j,n}$, $\Delta \mathbf{r}_j = \mathbf{r}_j - \mathbf{r}_{j,n}$, and $\Delta p_j = p_j - p_{j,n}$. Complete baseline equation (16) on a Dirichlet face by

$$\boldsymbol{\sigma}_{k,n+1}^{\text{tri}} = \boldsymbol{\sigma}_{k,n} + A_k [a_k(\Delta \mathbf{g}_k - \Delta \mathbf{u}_j) - R_k^n \Delta \mathbf{r}_j + \Delta p_j \mathbf{n}_k]. \quad (\text{D5})$$

The boundary-data increment is essential even though $\delta(\Delta \mathbf{g}_k) = 0$ during Newton differentiation.

Proposed one-sided plastic boundary completion. Use the same incremental cell correction as baseline equation (12):

$$\boxed{\boldsymbol{\sigma}_{k,n+1} = \boldsymbol{\sigma}_{k,n+1}^{\text{tri}} - A_k \overline{\mathbf{Q}}_j \mathbf{n}_k, \quad k \in I_{\mathcal{F}}^{\text{D}}.} \quad (\text{D6})$$

The subtraction in (D6) is a *new proposed numerical plastic boundary rule*. It is not a plasticity formula from the elastic TPSA paper. Together with baseline equation (17) for interior faces, it defines every traction row used in the residual. A Dirichlet reaction traction is unknown and generally has a nonzero Jacobian.

Boundary contributions to the retained auxiliary equations. For a prescribed face displacement, the retained elastic auxiliary fluxes are

$$\boldsymbol{\tau}_{k,n+1} = -A_k R_k^n \mathbf{g}_{k,n+1}, \quad v_{k,n+1} = A_k \mathbf{n}_k^T \mathbf{g}_{k,n+1}. \quad (\text{D7})$$

Let $\mathbf{b}_{r,j}$ and $b_{p,j}$ denote the integrated sources in the retained rotation and pressure equations. Their target-load right-hand sides in baseline equation (19) are

$$\begin{aligned} (\mathbf{f}_r)_j &= \mathbf{b}_{r,j} - \sum_{k \in \mathcal{N}_j \cap I_{\mathcal{F}}^{\text{D}}} \Delta_{j,k} \boldsymbol{\tau}_{k,n+1}, \\ (f_p)_j &= b_{p,j} - \sum_{k \in \mathcal{N}_j \cap I_{\mathcal{F}}^{\text{D}}} \Delta_{j,k} v_{k,n+1}. \end{aligned} \quad (\text{D8})$$

Thus these right-hand sides may change with the loading step while remaining fixed during its Newton iterations. Do not omit these boundary terms or add them a second time if the TPSA assembly already includes them.

For the full-Dirichlet reference discretization, the auxiliary blocks have the structure

$$\begin{aligned} A_{rr} &= -\text{diag}(V_j/\mu_j) \otimes I_{d_r}, \\ A_{pp} &= -\text{diag}(V_j/\lambda_j) - \Delta_{\text{int}} \text{diag}(A_k \delta_k^\mu) \Delta_{\text{int}}^*, \quad d_r = d(d-1)/2. \end{aligned} \quad (\text{D9})$$

In the second term, the diagonal contains interior-face entries only. Prescribed boundary displacements contribute to (D8), not to an additional unknown boundary-pressure stabilization term.

C ADDED: computing reconstruction and face Jacobian blocks

For a displacement column cell i , compute the auxiliary sensitivities by linear solves, with d right-hand sides per block:

$$A_{rr} \mathbf{R}_i = -(A_{ru})_i, \quad A_{pp} \mathbf{P}_i = -(A_{pu})_i. \quad (\text{D10})$$

Here $(A_{ru})_i$ and $(A_{pu})_i$ are the block columns for $\mathbf{u}_i$. Use the same constrained spaces as in the current auxiliary solve.

Define the condensed face-displacement derivative $G_{k,i}^c = \partial \mathbf{u}_k^c / \partial \mathbf{u}_i$. It is explicitly

$$G_{k,i}^c = \begin{cases} \Xi_{k,i} I_d - \delta_k^\mu \mathbf{n}_k (\Delta_k^* \mathbf{P}_i), & k \in I_{\mathcal{F}}^{\text{int}}, \\ 0, & k \in I_{\mathcal{F}}^{\text{D}}. \end{cases} \quad (\text{D11})$$

For nonadjacent i , $\Xi_{k,i} = 0$, but pressure condensation can still produce a nonzero interior $G_{k,i}^c$. For any $\mathbf{h} \in \mathbb{R}^d$, construct the strain derivative by substituting (D11) into the Green–Gauss sum:

$$\mathcal{B}_{j,i}^c \mathbf{h} = \frac{1}{2V_j} \sum_{k \in \mathcal{N}_j} A_k \left[(G_{k,i}^c \mathbf{h}) \otimes \mathbf{n}_{k,j} + \mathbf{n}_{k,j} \otimes (G_{k,i}^c \mathbf{h}) \right]. \quad (\text{D12})$$

Boundary faces contribute zero to this derivative because their data are fixed, although their displacement values contribute to the current strain. No direct rotation derivative is missing under the restricted scope (D3).

Interior traction blocks. Use baseline equations (23) and (24) with (D12). Both the trial-traction derivative and the plastic correction must include the pressure sensitivity. The local tensor $\mathbb{C}_j^e - \mathbb{C}_j^{\text{alg}}$ already includes multiplier and flow-direction variations.

Dirichlet traction blocks. Let $(\mathbf{R}_i)_j$ extract the rotation rows for the adjacent cell j , and $(\mathbf{P}_i)_j$ its pressure row. Let δ_{ji} be the cell Kronecker delta, distinct from the distance δ_k^j . Differentiating (D5)–(D6) gives, in 3D,

$$\mathbf{J}_{k,i} \mathbf{h} = A_k \left[-a_k \delta_{ji} \mathbf{h} - R_k^n (\mathbf{R}_i)_j \mathbf{h} + \mathbf{n}_k ((\mathbf{P}_i)_j \mathbf{h}) - ((\mathbb{C}_j^e - \mathbb{C}_j^{\text{alg}}) : (\mathcal{B}_{j,i}^c \mathbf{h})) \mathbf{n}_k \right], \quad k \in I_{\mathcal{F}}^{\text{D}}. \quad (\text{D13})$$

The prescribed face displacement has zero derivative, but the adjacent cell displacement, pressure, rotation, and plastic correction do not.

From an action to a matrix. For Cartesian basis vectors $\mathbf{e}_\ell$, form all columns:

$$(\mathbf{J}_{k,i})_{;\ell} = \mathbf{J}_{k,i} \mathbf{e}_\ell, \quad \ell = 1, \dots, d. \quad (\text{D14})$$

Use the interior law for interior faces and (D13) for Dirichlet faces. Then assemble all residual-cell/displacement-cell blocks with baseline equation (25): $\mathbf{J}_{a,i}^{\text{glob}} = \sum_{k \in \mathcal{N}_a} \Delta_{a,k} \mathbf{J}_{k,i}$. Include Dirichlet reaction rows. There is no extra face-area factor.

D ADDED: initialization and complete computation order

Committed state and initialization. Store $\mathbf{u}_n, \mathbf{r}_n, \mathbf{p}_n$, the old boundary data $\mathbf{g}_n$, the reconstructed strains $\boldsymbol{\varepsilon}_{j,n}^{\text{rec}}$, constitutive histories $\mathcal{H}_{j,n}$, and numerical face tractions $\boldsymbol{\sigma}_{k,n}$. The old strain must use the old boundary data, not the target $\mathbf{g}_{n+1}$. Cell stresses satisfy $\boldsymbol{\sigma}_{j,n} = \mathbb{C}_j^e : (\boldsymbol{\varepsilon}_{j,n}^{\text{rec}} - \boldsymbol{\varepsilon}_{j,n}^p)$. Numerical face tractions are stored from the same numerical flux law; do not reconstruct them using an unrelated cell-stress average.

For an undeformed initial state with zero initial displacement data, zero initial sources, and an admissible unstressed material, a simple consistent initialization is $\mathbf{u}_0 = \mathbf{r}_0 = \mathbf{p}_0 = 0$, $\boldsymbol{\varepsilon}_{j,0}^{\text{rec}} = \boldsymbol{\varepsilon}_{j,0}^p = 0$, $\kappa_{j,0} = 0$, and $\boldsymbol{\sigma}_{k,0} = 0$. Nonzero initial states require supplied compatible history, boundary data, and numerical face tractions.

Step	Computation at load step $n \rightarrow n + 1$
0	Specify the full Dirichlet data, material routine, geometry, and fixed interior weights. Assemble the reference TPSA operators with their boundary terms. Prepare constrained auxiliary solves.
1	Choose the current cell displacement $\mathbf{u}$. Solve baseline (19) for $\mathbf{r}, \mathbf{p}$ using the target right-hand sides (D8).
2	Reconstruct interior face displacements using (3); set every boundary face displacement using (D1). Compute total cell strains using (4).
3	Evaluate every return map from the unchanged $\mathcal{H}_{j,n}$. Obtain the returned stress, history candidate, incremental $\bar{\mathbf{Q}}_j$, and matching $\mathbb{C}_j^{\text{alg}}$.
4	Form interior tractions using (17) and Dirichlet tractions using (D5)–(D6). Assemble the current residual using (8).
5	Compute or apply $\mathbf{R}_i, \mathbf{P}_i$, then $G_{k,i}^c$ and $\mathcal{B}_{j,i}^c$. Construct all interior and Dirichlet face Jacobian blocks and assemble with (25).
6	Solve $\mathbf{J}^{\text{glob}} \delta \mathbf{u} = -\text{res}^c$, update the cell displacement, and repeat steps 1–5 until convergence. Keep $\mathbf{g}_{n+1}$ and committed history fixed throughout.
7	After convergence, commit the candidate material states, converged cell variables and reconstructed strains, target boundary data, and all numerical face tractions.

The fixed linear sensitivities and trial-traction derivatives may be prepared once and reused. Current return maps and algorithmic tangents must still be evaluated at each Newton iterate. No explicit dense inverse is required. Full face Dirichlet conditions leave the cell displacement unknowns in the system and remove the usual displacement rigid motions.

Pressure solvability. For finite positive λ_j , use the actual pressure block (D9). In the incompressible reference limit on a connected domain, impose a pressure gauge and ensure compatibility of the retained pressure equation:

$$\sum_{k \in I_{\mathcal{F}}^D} A_k \mathbf{g}_{k,n+1} \cdot \mathbf{n}_k = \sum_j b_{p,j}. \quad (\text{D15})$$

For zero volumetric source the prescribed boundary flux must sum to zero. Gauge fixing alone does not repair incompatible boundary data. Apply the condition and gauge per connected component when necessary. This limit statement concerns the retained pressure equation; the local material update above uses finite elastic coefficients. An exact incompressible constitutive model requires its own material integration treatment.

Derivative verification. When perturbing cell displacement to check the Jacobian, keep target Dirichlet data, weights, and $\mathcal{H}_n$ unchanged; recompute auxiliaries, strains, and return maps. Check interior and boundary traction derivatives as well as the assembled residual on a fixed smooth constitutive branch. These additions complete the computation of the specified proposed scheme; they do not establish its spatial convergence.